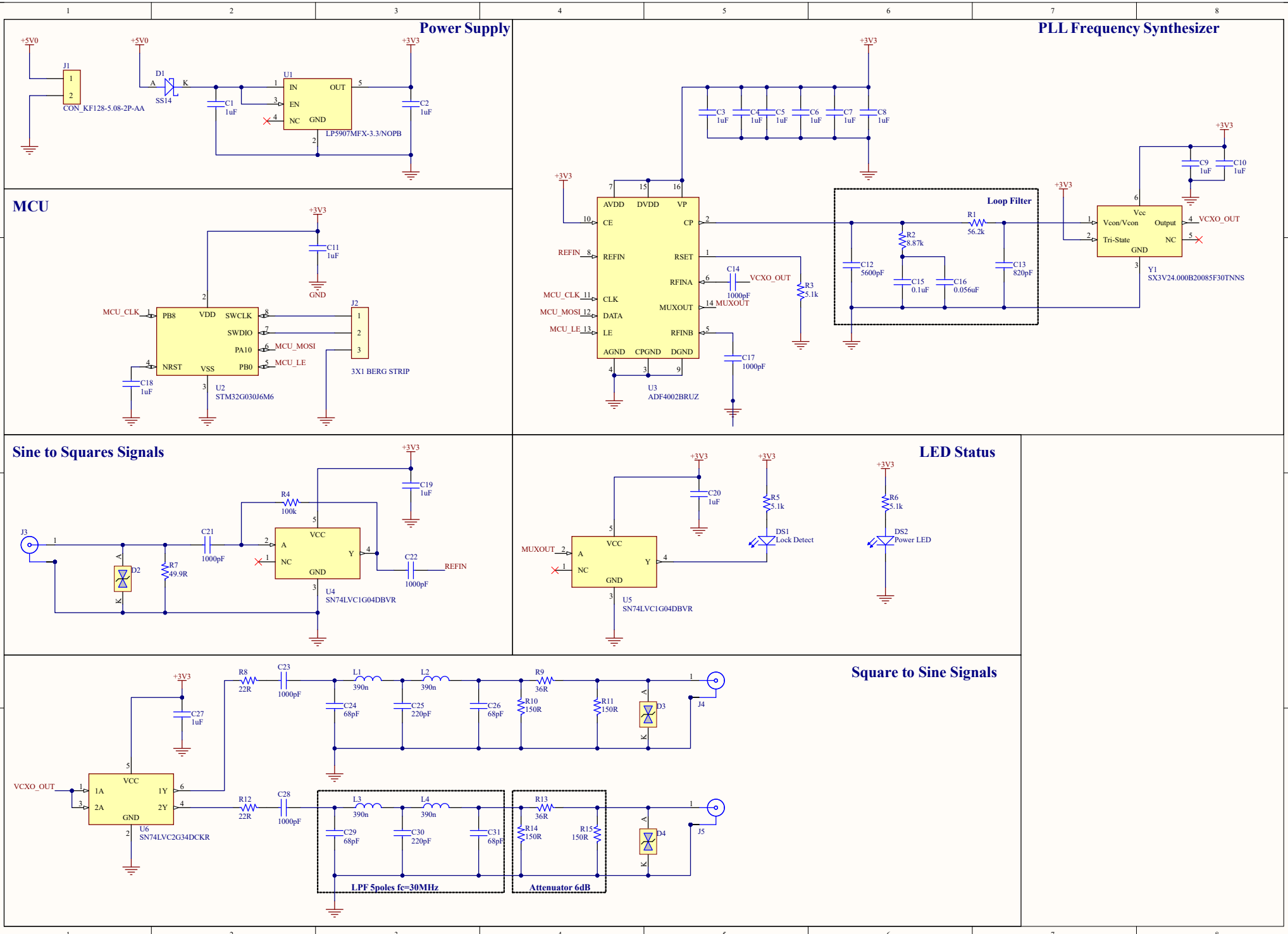

Power Supply
+5V0
J1
CON_KF128-5.08-2P-AA
D1
SS14
C1
1uF
U1
IN
OUT
EN
NC
GND
LP5907MFX-3.3/NOPB
C2
1uF
+3V3
MCU
C11
1uF
MCU_CLK
PB8
VDD
SWCLK
SWDIO
PA10
NRST
VSS
PB0
MCU_MOSI
MCU_LE
C18
1uF
U2
STM32G030J6M6
J2
3X1 BERG STRIP
PLL Frequency Synthesizer
C3
1uF
C4
1uF
C5
1uF
C6
1uF
C7
1uF
C8
1uF
AVDD
DVDD
VP
CE
CP
REFIN
RSET
RFINA
CLK
MUXOUT
DATA
LE
RFINB
AGND
CPGND
DGND
C14
1000pF
VCXO_OUT
MUXOUT
R3
5.1k
C17
1000pF
U3
ADF4002BRUZ
Loop Filter
R1
56.2k
R2
8.87k
C12
5600pF
C15
0.1uF
C16
0.056uF
C13
820pF
Vcc
Vcon/Vcon
Output
Tri-State
NC
GND
C9
1uF
C10
1uF
Y1
SX3V24.000B20085F30TNNS
Sine to Squares Signals
J3
C21
1000pF
R4
100k
C19
1uF
D2
R7
49.9R
VCC
A
NC
Y
GND
C22
1000pF
REFIN
U4
SN74LVC1G04DBVR
LED Status
C20
1uF
R5
5.1k
DS1
Lock Detect
R6
5.1k
DS2
Power LED
U5
SN74LVC1G04DBVR
Square to Sine Signals
C27
1uF
R8
22R
C23
1000pF
L1
390n
L2
390n
R9
36R
C24
68pF
C25
220pF
C26
68pF
R10
150R
R11
150R
D3
J4
VCXO_OUT
VCC
1A
1Y
2A
2Y
GND
U6
SN74LVC2G34DCKR
R12
22R
C28
1000pF
L3
390n
L4
390n
R13
36R
C29
68pF
C30
220pF
C31
68pF
R14
150R
R15
150R
D4
J5
LPF 5poles fc=30MHz
Attenuator 6dB